

Qatar-2b

R-0976

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-2_250627 · created 2026-07-29T18:54:40+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460853.7266 +/- 0.0019 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.167 +/- 0.065
Transit depth (Rp/Rs)^2	2.79 +/- 2.16 [%]
Orbital Inclination (inc)	89.07 +/- 1.69
Airmass coefficient 1 (a1)	4.25 +/- 1.29
Airmass coefficient 2 (a2)	-0.767 +/- 0.057
Scatter in the residuals of the lightcurve fit is	25.47 %
Best Comparison Star	None
Optimal Aperture	0.75
Optimal Annulus	6.75
Transit Duration (day)	0.0762 +/- 0.0047

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.167 ± 0.065 vs 0.16327 (deviation 0.03σ)
Duration fit vs published	0.0762 d vs 0.0767 d (deviation 0.06σ)
Mid-time O–C	-4.7 min
Depth SNR	1.5
Residual scatter	25.47 %

Light curve

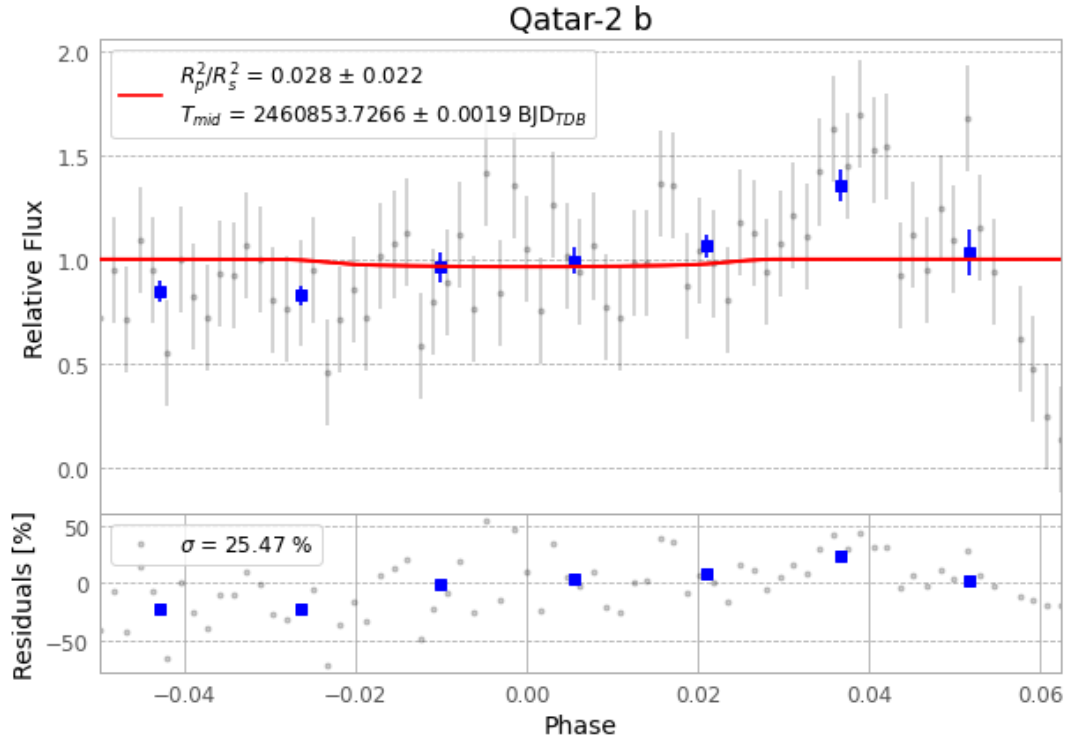


Figure 1 — FinalLightCurve_Qatar-2 b_2025-06-26.png (SHA-256 1a892c71d9ba5e9a...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [461, 430], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 ba70d7e6319119a2...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

73 FITS frames (48 MB), Qatar-2250627034515.FITS → Qatar-2250627072115.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-2-250627.log (53a4149cca33...), FinalLightCurve_Qatar-2 b_2025-06-26.png (1a892c71d9ba...), FinalLightCurve_Qatar-2 b_2025-06-26.csv (9e987e3787e0...)

Compute resources

Wall time 105.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-29 18:54Z.